

‘Maize is Life, but Rice is Money!’ A Village Case Study of the 2001/02 Famine in Malawi

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This paper provides a case study of the 2001/02 famine in Malawi from a village in the Southern Region of the country. Based on in-depth micro-level field research, it challenges some commonly accepted views about this crisis. The paper provides evidence that: (1) there was a serious ‘famine’ in the community; (2) the decline in food availability was not the major causal factor of the famine; (3) the early warning system in the rural areas was functioning appropriately and the famine did not happen in ‘silence’, unnoticed; (4) the food preferences of Malawians are not ‘inflexible’; and (5) the famine, contrary to the claims of some of the ‘new famine hypotheses’, was less the consequence of underlying vulnerability and long-term social or economic trends but, rather, the result of an unexpected and sudden shock, which was generated by the exponential increase in the price of all food crops.

Keywords: famine, Malawi, case study

INTRODUCTION

This paper provides a case study of the Malawi famine of 2001/02 from a village located in one of the worst-affected areas in the Southern Region of the country. The study builds on six months of fieldwork, during which village-level data were collected and victims of famine were interviewed. Because it builds from the grassroots level, this paper provides a different view of the crisis as opposed to macro-level analyses and can therefore challenge commonly accepted views.

There are at least three reasons that justify a detailed case study analysis of this famine. First, in spite of generating extraordinary attention in academic and public debate, a great deal remains unknown about the causes of the crisis. Second, the severity of the famine remains subject to debate, with widely varying estimates of famine mortality. Third, several new hypotheses have emerged to explain the causes of this famine, which have advanced the theoretical debates about the general causes of famines. These three characteristics of the famine are discussed briefly in the following paragraphs.

Various studies have documented the famine at the national level, most of which have left several questions unexplored (see Devereux 2002; Devereux and Tiba 2007). Among these was the rather controversial decision that, before the famine hit, the fully stocked national Strategic Grain Reserve was sold by the government. Not only was the maize sold at the wrong time, before the new harvest fell by 32 per cent in June 2001, but no buffer stock was retained for an emergency. During the crisis the price of maize increased exponentially, which was a clear

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indicator of excess demand for maize on the market. The government parastatal ADMARC (Agricultural Development and Marketing Corporation), which had been selling maize at a stable price of MK5 per kilogram (MK = Malawi Kwacha) for several years before the famine, having no access to maize from the reserves and with no (or an insignificant amount of official) imports arriving in the country, failed to intervene and defend maize prices even after raising the price of maize to MK17. Research by the Anti-Corruption Bureau revealed that officials ‘corruptly benefited from the purchase of maize from the Strategic Grain Reserve’ (Government of Malawi 2002b, 1) and it is known from export statistics that ‘the bulk of maize in the Strategic Grain Reserve was sold locally and was available for local consumption in 2001’ (Rubey 2004, 1). The mismanagement of the national reserves may provide evidence of the existence of a ‘controlling institution’ in the country; that is, a set of overlapping interests between private and government trade, which have the means to control and regulate the maize market in the country (Tiba 2005, 80).

During the crisis, decision-makers repeatedly denied that a famine was happening or even developing in the country, by claiming that according to the official production figures of root and tuber crops, there was enough cassava and sweet potato available for consumption in the country (Devereux 2002, 6). This led to the argument that the food preferences of Malawians are ‘inflexible’; however, the reliability of these figures, which will be examined later on, was not questioned or investigated. The government only declared a ‘State of Disaster’ (but not a *famine*) on 27 February 2002, by which time the famine was almost over, as a result of the new incoming harvest in April.

The severity of the crisis also remains subject to debate, as indicated by the lack of consensus on whether it should be labelled a ‘famine’. There is no agreed definition in the literature on which to base judgement (Howe 2002, 20), but the use of the word *famine* would suggest a more severe crisis. Differing commentators have labelled the crisis a ‘State of Disaster’ (Devereux 2002), a ‘Food Crisis’ (Stambuli 2002; Devereux et al. 2003; Dorward and Kydd 2004) or a ‘Maize Crisis’ (FEWS, October 2001), while others have explicitly referred to it as a *famine* (ActionAid 2002; Owusu and Ng’ambi 2003; Bryceson 2006; Peters 2006, 325; Devereux and Tiba 2007). There remain great variations in the estimates of hunger-related mortality, ranging from 300 (Taifour 2002, 2) to 46,000–85,000 deaths (Devereux and Tiba 2007, 145). Although even the lowest estimate of mortality is higher than the 200 people who died during the Nyasaland *famine* in 1949 (Vaughan 1987, 162), the ‘government and the donor community [. . .] continue to deny that a famine occurred in Malawi in 2001–2002’ (Devereux and Tiba 2007, 162). In this paper, I refer to the crisis as a *famine*, this being substantiated by interviews with those who lived through the crisis, whose views are ‘worth incomparably more than those of textbooks and dictionaries’ (Devereux 1993, 19).

In addition to the above *practical* debates, it is worth exploring some recent changes in the *theoretical* literature on famines. The traditional approach to causes of famine has been based on the presumption that decline in aggregate food availability is a necessary condition for famines to happen, which ‘implies a strong and constantly operating check on population from the difficulty of subsistence’ (Malthus 1798, 5). In *Poverty and Famines*, Sen argued that famines *can* happen without a decline in aggregate food availability, and using the example of four famines he rebutted what he called the food availability decline theory. In his ‘entitlement approach’, he shifted the analysis from aggregate macro-level food availability to individual access to food (Sen 1981).¹ The ‘Food Availability Decline’ and ‘Food Entitlement Decline’ theories have been the most widely used analytical frameworks to investigate the causes of famines.

¹ For an overview of the literature on entitlement, see Fine (1997) and Osmani (1995).

At least three new ideas have been used to analyse the Malawi crisis. First, the 'New Variant Famine' (NVF) hypothesis has explained the famine by reference to the HIV/AIDS pandemic. The underlying argument is that while in previous 'traditional' famines mainly the young and the elderly have suffered the most, in an environment characterized by HIV/AIDS, the sexually active working-age population is affected worst. This restructures household income and expenditure patterns, dramatically impacts coping strategies, and creates a uniquely vulnerable environment in which the 'prospects for a sharp decline into severe famine are increased, and the possibilities for recovery reduced' (de Waal 2003; see also de Waal and Whiteside 2003).

Others have argued that the famine has been the consequence of liberalization policies implemented in Malawi during the past two decades. The structural adjustment programmes have affected food security directly through market liberalization, price deregulation and the removal of subsidies on agricultural inputs, which has resulted in rising prices (Chilowa 1998; ActionAid 2002). One of the most radical critiques is offered by Owusu and N'gambi, who stress that Malawi's 'food security infrastructure has been *torn apart* by IMF-led economic liberalization' and 'to a large extent, the liberalization and privatisation policies of the IMF/World Bank on Malawi *are to blame* for the food insecurity' (Owusu and N'gambi 2002, 20, 33; emphasis added). At the same time, risks of opportunistic behaviour, such as corruption in the state sector, and coordination failure have emerged, as a weak regulatory framework for liberalization has resulted in private companies using their resources to 'influence decision makers and create dominant market positions' (ibid., 38; see also Dorward and Kydd 2004).

Finally, the Malawi famine becomes particularly interesting in light of the argument that 'no substantial famine has ever occurred in any independent country with a democratic form of government and a relatively free press' (Sen 1999, 152). After gaining independence in 1964, Malawi was a dictatorship for three decades, ruled by the life president Dr Hastings Kamuzu Banda. During this period, both political and economic power was heavily concentrated in the hands of the president and his narrow political elite. In spite of this, there was no famine or hunger-related mortality recorded in the country during this period. The last recorded famine happened in 1949, when Nyasaland (Malawi) was a British Protectorate: as an independent country, Malawi had never experienced a serious famine. It is therefore difficult to apply Sen's argument in the context of Malawi: for three decades, a centrally regulated agricultural sector under a strictly enforced dictatorship proved successful in averting a major food crisis, while after less than a decade of political and press freedom, this economy experienced famine.

All of this has raised several questions for the purpose of a village case study. First, how severe was this crisis and would it qualify as a 'famine'? Second, was the early warning system functioning and was there any documented evidence of a famine in the area? Third, were roots, tubers and other food crops indeed available during the famine, and are the food preferences of Malawians really so 'inflexible' that people would refuse to consume any food other than maize during such a crisis? Fourth, would it have been possible to predict a market failure and the increase in the price of maize, and how did the local markets, including ADMARC, function during the crisis?

The structure of the paper is as follows: the first section describes the methodology used during the fieldwork and explains the selection of the area. The characteristics of the village and livelihoods in the area are explored in the second section. This is followed by a section presenting evidence of a famine in the village. The fourth section investigates trends and 'pre-famine conditions', and is followed by a section that investigates the causes of the crisis. Finally, the implications for future research are noted and the analytical conclusions are summarised.

METHODOLOGY AND AREA SELECTION

The fieldwork for this study was conducted in 2003, over a period of six months between January and June. Qualitative and quantitative research techniques were combined to gather information. The qualitative techniques included key informant interviews, focus group discussions, household case studies, transect walks, mapping and participant observation (Joergensen 1989; Bryman 2001). In addition, information was collected from local institutions in the area and from households that were located within the boundaries of the village, by using a household questionnaire.² The questionnaire had five sections, focusing on household demography, poverty and ownership of assets, food production and sources of food, the impact of the famine, and food aid receipts. All data gathered in the questionnaire were analysed using SPSS software.

For the purpose of the research, the village chosen was within Zomba District in Lake Chilwa basin in the Southern Region of Malawi. The following selection criteria were used: underlying indicators of vulnerability (especially the prevalence of HIV/AIDS and the high population density) and food insecurity (amongst the highest in southern Malawi, a region of the country badly affected by the famine; see Devereux et al. 2003). The village is located next to a river where floods destroyed maize production during 2001, but the unique feature of the area is that rice is produced in addition to maize, and rice production did *not* decline before the crisis. It is estimated that about 50 per cent of the rice harvested in Malawi comes from the Lake Chilwa wetland area (Maloya 2001). These characteristics made it possible to investigate a number of apparent paradoxes, such as: (1) why the village experienced a famine during the 2001/02 season despite the rice production; (2) to what extent 'inflexible food preferences' should be regarded as causing the crisis (if rice was available); and (3) whether the maize and rice markets contributed to the crisis.

VILLAGE CHARACTERISTICS AND LIVELIHOODS

Namasalima village is located in the Southern Region of Malawi in Zomba District, between Zomba town and Lake Chilwa, 35 km away from the town and 10 km from the lake. Due to favourable agro-ecological conditions in the lake basin and the availability of water from the Domasi river, the dominant crop in the area is rice. An irrigated rice scheme (the Domasi Rice Scheme) of 470 ha was established by the government in the early 1970s on the other side of the river, which provides access to irrigated plots for 1,305 farming families in the area as well as opportunities for casual work (*ganyu*). Because the Scheme is irrigated, it is possible to produce winter crops (both maize and rice) during the dry season, for which a (relatively high) rental fee must be paid to the Association (Ferguson and Mulwafu 2004). The most important institutions in the village include the Health Centre, which has a catchment area of 43 villages, the local primary school and the government market (ADMARC). ADMARC is located on the other side of the river (approximately 3 km away from the centre of the village) and it supplies a large area and population with maize. The next ADMARC market is 25 km away from the village (Malosa).³

² This included 256 households in total. The decision not to sample, but to investigate all households, was driven by the second objective of the study – to analyse the distribution of food aid in the village, which was often shared among households; therefore, selecting a few 'beneficiary' households would have been problematic.

³ Occasionally, seasonal ADMARC centres are opened in the area, but only on a temporary basis; they also sell much less maize than the regular market.

Table 1. Poverty in the village, as measured by possession of household items

<i>Household owns</i>	<i>Rural average</i>	<i>Namasalima village</i>
Radio	58.5	50.8
Bicycle	41.8	45.8
Pit latrine	80.0	78.0
Main floor is earth	87.1	89.1
Main floor is cement	12.0	10.9

Source: National Statistical Office (2004, 22–4) and Namasalima Survey, 2003.

Figures are percentages.

The area suffers from food insecurity for several reasons. First, the river often floods the village(s) during the rainy season, due to excessive rainfall and environmental degradation, and because the dyke on the northern side of the river, built to protect the Domasi Irrigation Scheme, makes the southern side vulnerable to floods. Second, over the past 25 years, the population density in the Traditional Authority (TA)⁴ has grown at a rate nearly 75 per cent above the national average and the average per capita land holding size has fallen from 0.9 to 0.51 ha (Tiba 2005, 362). As a consequence of recurring floods and population pressure on the land, the fertility of the soil has become very poor. Third, the 16 km long dirt road that connects the village to the tarmac becomes impassable during the rainy season, making transportation and commuting in the area difficult. Finally, there are no public forest reserves or wild fruit trees in close vicinity to the village: the nearest reserves are in the Chikala Mountains, about 30 km away. As shown in Table 1, the poverty level in the village, measured by the characteristics of households and the possession of items, appears to be very similar to that of other rural areas in Malawi.

Rice and casual labour are thus the two most important sources of income in this village, and the subsequent analysis about the causes of the famine will therefore put substantial emphasis on investigating trends in these two sources of income.

INDICATORS OF FAMINE

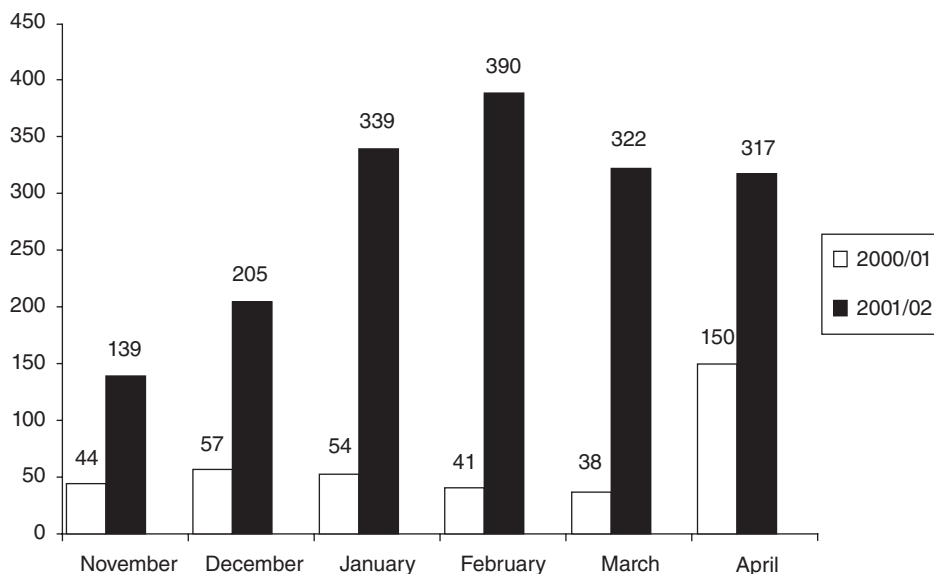
Whether there was a *famine* and how severe the crisis was in the village can be assessed by examining several indicators. During the fieldwork, quantitative data on child malnutrition, morbidity, school enrolment and mortality were collected, as well as qualitative information in informal interviews with the households and individuals affected by the famine. These sources allow an assessment of the impact of the crisis from different angles.

The primary source monitoring malnutrition rates of children under the age of five is the Namasalima Health Centre, which collects monthly data from the 43 villages in its catchment area.⁵ Figure 1 compares the number of malnourished children diagnosed at the Health Centre during the famine with the number of malnourished children the year before, showing that in

⁴ The area is divided by different administrative boundaries. The Kumtumanji Traditional Authority (TA) shares the same boundary with the Nsondole Environmental Planning Area (EPA), as used by the Ministry of Agriculture. On the other hand, the Namasalima Health Centre has a different catchment area, which extends beyond the TA and EPA boundaries.

⁵ At the time of the survey, there were 43 villages in the area, containing 19,752 people (of whom 3,299 were children under the age of five) living in 4,323 households (Namasalima Health Centre data).

Figure 1 Under-five malnutrition at Namasalima Health Centre, 2000–2002: the units are the number of malnourished children diagnosed and recorded at Namasalima Health Centre



late 2001 child malnutrition increased dramatically, peaking in February, when it was almost 10 times higher than the year before. The relative increase in child malnutrition is a clear indication of a severe shock to the community.⁶

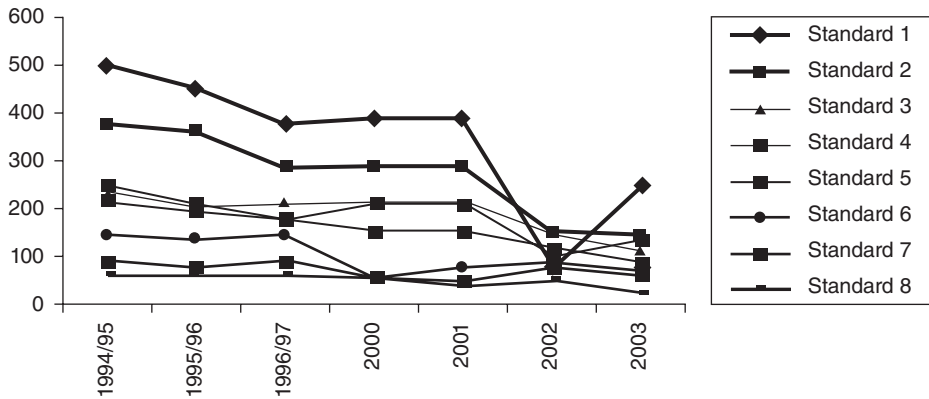
The prevalence of cholera during the famine also indicates severe social breakdown in the village.⁷ Apart from a mild cholera outbreak in 1999 (with 12 registered cases), no other cases of cholera had been detected by the Health Centre in any of the years since 1998. During the famine, however, 90 cases were recorded between November and March. This coincided with a national crisis, as 'the majority of districts are experiencing other health problems [. . .] 23 districts are experiencing a cholera outbreak with a total of 18,026 cases reported and 503 deaths' (Government of Malawi 2002c). Although the Health Centre does not collect information on hunger-related diseases, the household questionnaire indicates that 9.2 per cent of the village population suffered from a hunger-related disease during the crisis.⁸

⁶ A child is considered wasted if his weight for height (WH) measurement is -2 SD below the reference score. Wasting is an indicator of acute or recent nutritional deficits and is closely tied to risk of mortality. However, anthropometric measurements show a wide range of food deprivation and are not just narrowly related to energy intake. The prevalence of malnutrition in the region may well have been higher during the famine than the figures indicate, because the figures only include those children who had attended the Health Centre.

⁷ Cholera is an acute, diarrhoeal illness, caused by infection of the intestine. It is not a hunger-related disease, but it is associated with deteriorating hygienic conditions. The cholera cases diagnosed by the health centre may have underestimated the real prevalence of the disease in the area, as many of the people already affected by cholera may not have reached the health centre in time.

⁸ The questionnaire collected information on the occurrence of other malnutrition diseases, such as oedema (for adults) and kwashiorkor and marasmus (for children) in surveyed households. All of these diseases are related to hunger and are caused by inadequate food intake (see Latham 1997). It is unlikely that respondents were able to distinguish between the symptoms and identify which of these diseases they had suffered from, but they were clearly able to indicate whether they had experienced a hunger-related disease in the household.

Figure 2 School enrolment rates in Namasalima Primary School, 1994–2003 (Source: Namasalima Primary School, 2003)

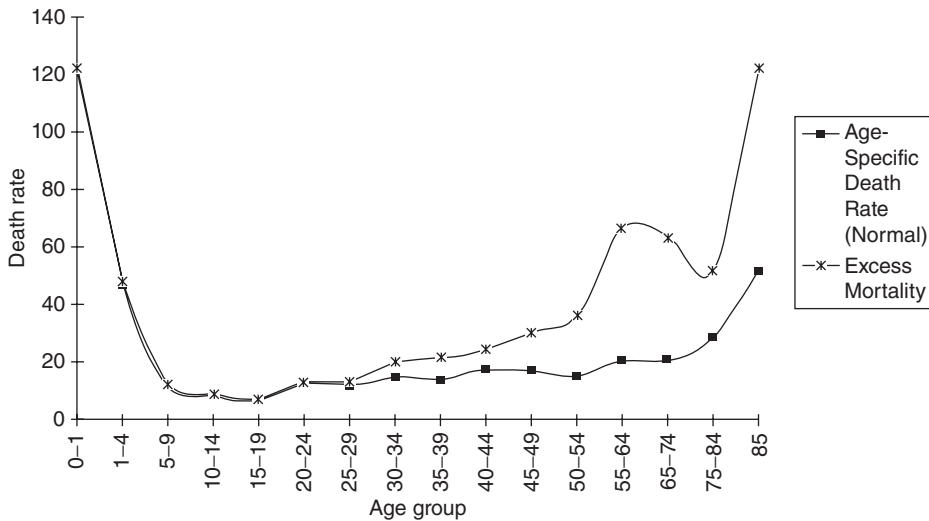


In addition to health indicators, figures from the local primary school also show severe social breakdown in the village during the crisis (Figure 2). The average number of pupils enrolling in school at the beginning of the academic year was 1,589, for the period between 1994 and 2001, fluctuating between 1,869 and 1,420 students. In January 2002, however, only 811 students started their studies and, as the figure below indicates, the sharpest fall in enrolment was evident in the two lowest classes. In 2001, 385 pupils enrolled in Standard 1. In contrast, in January 2002, only 75 enrolled; that is, there was a shortfall of 80 per cent of the 'expected enrolment'.⁹ In normal years, only 2 per cent of children drop out of school after attending Standard 1 (National Statistical Office 2004, 19, table 2.6). The headmaster of the school in Namasalima confirmed that there was no other reason for the low enrolment apart from the famine in early 2002. Indeed, in the following year enrolment returned to its 'normal' level. Another shocking indicator is the number of primary school completers who failed the national examination in November 2002. While, during the five years before the crisis, on average, 48–70 per cent of the students finishing primary school passed the final examination, in 2002 as many as 93 per cent of them failed. The headmaster of the school noted, 'pupils were not coming to school between January and March 2002 because of hunger. If a student does not attend school for three months, what would be the result of the exam?'

There are other indicators of social tensions in the village during January and February 2002, which have been derived from the household questionnaire. Thirty-nine people were attacked in the village while transporting food and 51 per cent of the interviewed households reported that they felt it necessary to take precautionary measures to protect their fields. People resorted to staying in their fields overnight to guard crops and resorted to storing food in their houses. Those caught stealing were publicly beaten on the main street of the village and, in one case, a person was killed in one of the surrounding villages for having stolen maize from a field. The crisis was commonly regarded as 'worse than the famine of 1949', which also hit Zomba District, mainly because, as one of the key informants noted, 'there were more people affected'.

⁹ Another study conducted in a local primary school in Mchinji District found a similar impact on school attendance. In that school, the attendance rate during the famine dropped to 'less than 20 per cent', which tallies with our findings from Namasalima (Munthali 2006, 39).

Figure 3 Baseline and excess (famine) mortality in 43 villages between October 2001 and March 2002: the units are the death rates per 1,000 mid-year population



As the population has grown more than fivefold since 1949 (National Statistical Office 1998a), it could be argued that if the famine in 2002 was half as severe as the famine of 1949, there would be 2.5 times more people affected in the area.

In qualitative interviews, respondents frequently referred to the fact that ‘deaths, aggravated if not entirely caused by hunger, also occurred among families’, echoing the findings of other research in neighbouring villages (Ferguson and Mulwafu 2004, 10). To follow up on this argument, figures on hunger-related mortality were collected from the catchment area of the Health Centre, covering 19,732 people living in 43 villages. Village chiefs and key informants were asked to convene meetings and record the name, age, sex and main activity of people who died of hunger in their village during the crisis. According to the survey, 122 people died of hunger in the Namasalima Health Centre Catchment Area between October 2001 and March 2002. It is likely that these deaths were hunger-related: 91 per cent of total mortality (111 deaths) occurred between January and March, and half of all these recorded deaths took place during the peak of the hungry season in February, when the prices of all food crops sky-rocketed (see later). Figure 3 compares *excess* mortality rates, defined as the proportion of the population within the respective age cohort who died of hunger according to the survey, with the *baseline* Age Specific Death Rates (ASDR), as reported by the last Population and Housing Census (National Statistical Office 1998a).¹⁰

The figure shows that hunger-related mortality began to increase only after the age of 30, and that 62 per cent of the total number of people who died of hunger were over 50 years old. There were very few excess deaths occurring below the age of 30, despite the dramatic increase in child malnutrition and the significant fall in school enrolment. One possible explanation for the absence of excess deaths reported for the 0–5 age group is that death rates below the age of five are so high in normal years, that even if hunger was a contributory factor to a child’s

¹⁰ The Population and Housing Census reports the Baseline Age Specific Death Rate for Zomba district, in which the age cohorts are as shown in Figure 3 (National Statistical Office 1998a, 104, table 7.2).

death, it would not be considered 'famine mortality' by the community. Therefore, people in the Namasalima area would not consider the death of a young child as an abnormal or, ironically speaking, a 'surprising' event. This explanation was repeatedly offered in qualitative interviews and focus group discussions. The reason why other (young) age groups are not represented in these figures is less clear, but one explanation could be that having survived infancy, older children have an inherent ability to recover from disease. This finding is consistent with Devereux's argument that 'the famine was known as "the swelling", because malnourished adults suffered oedema that caused their lower legs to swell up shortly before they died' (Devereux 2002, 19), but it apparently contradicts demographic evidence from other famines, in which 'young children . . . and the elderly experience the greatest absolute increases in death rates' (Dyson and Ó Gráda 2002, 10).

These results can be compared with different benchmarks and thresholds, as well as with other surveys of mortality carried out in Malawi during the same period between January and March 2002. Although there is no consensus in the humanitarian community about the Crude Death Rate (CDR) that should be considered as evidence of an emergency or a famine, it is usually accepted that a CDR of 1 per 10,000 per day is a threshold marking the 'emergency phase of a complex humanitarian emergency' (Spiegel et al. 2002, 1,927). Salama et al. (2001) have used the same benchmark to argue that a famine occurred in Ethiopia in 2000. In their 'famine intensity scales', Howe and Devereux (2004, 248) use the same cut-off to classify a social process as a famine.¹¹ The CDR calculated from the Namasalima survey for February is 1.8227 per 10,000 per day, which would certainly qualify as a *famine* according to the conventional benchmarks. It is remarkably close (only 7.5 per cent lower) to the Crude Death Rate of 1.96 reported by WHO for the same period (WHO 2002, 5).

It is possible to extrapolate these figures to the national level and provide an estimate of total famine mortality in Malawi during 2001/02. Based on the Excess Age Specific Death Rates derived from the survey, the national excess mortality due to famine would amount to 65,279 deaths. Out of these, 5,628 people would have been under the age of 19, 19,134 between the ages of 20 and 50, and 40,518 people over the age of 50 would have died of hunger during the famine. This figure falls within the range of the Malawi Food Crisis Impact Survey, which estimated that famine-related mortality was within the range of 46,000–85,000 deaths (Devereux and Tiba 2007, 145). WHO Malawi and the Ministry of Health also conducted a rapid assessment of the health impact of the crisis during April 2002 and estimated the Crude Death Rate during the crisis. Extrapolating their estimates of famine mortality to the national level indicates that total famine deaths would amount to 80,360, but this figure should be seen as an upper estimate, because the study was carried out in the worst-affected districts of the country (ibid.).

The limitations of these figures should be acknowledged. First, there is no way to verify *ex post* whether an individual recorded in the questionnaire died of hunger, of a hunger-related disease or for any other reason; in other words, whether or not the reported mortality, even if reliable, should be considered *excess* (famine mortality) and not part of, and lying within, the average 'normal' fluctuations in the mortality rate.¹² It may also be argued that village chiefs had

¹¹ Howe and Devereux (2004, 248) distinguish between food insecurity ($0.2 < \text{CDR} < 0.5$), food crisis ($0.5 < \text{CDR} < 1$), famine ($1 < \text{CDR} < 5$), severe famine ($5 < \text{CDR} < 15$) and extreme famine ($\text{CDR} > 15$) conditions, depending on the CDR and/or wasting.

¹² The prevalence of other, not strictly hunger-related, diseases (cholera) increased substantially during the famine, which also had an impact on regional mortality figures, but the contribution of other diseases to the mortality figures cannot be isolated.

an interest in over-reporting the number of hunger-related deaths in their village (in anticipation of future assistance), despite the fact that it was made clear that the data were being collected for purely scientific purposes. Third, data should be interpreted with caution because our findings are based on a relatively small sample, which is further broken down into age subgroups (and, in what follows, by months). Consequently, even the smallest change in the figures can cause a large change in the overall percentages. There may be other problems with our calculations, which assume that mortality in 'normal years' occurs with the same frequency in every month during the year. The nutritional intake of the poor and ultra-poor falls in the 'hungry months', which may increase their risk of dying even in normal years; and the prevalence of malaria (as well as other diseases such as diarrhoea or pneumonia) also increase during the rainy season. The extrapolation of figures is based on the assumption that this region is representative of other districts of Malawi, which may not be the case. In spite of all of these limitations, our figures do fall within the range of two other independent surveys conducted over the same period in Malawi. More importantly, the data provide ample evidence that previous estimates of famine mortality, such as 300–500 deaths (Taifour 2002, 2) or 1,000–3,000 deaths (Catholic Commission for Justice and Peace, cited in Devereux 2002, 18), need to be revised upwards significantly.

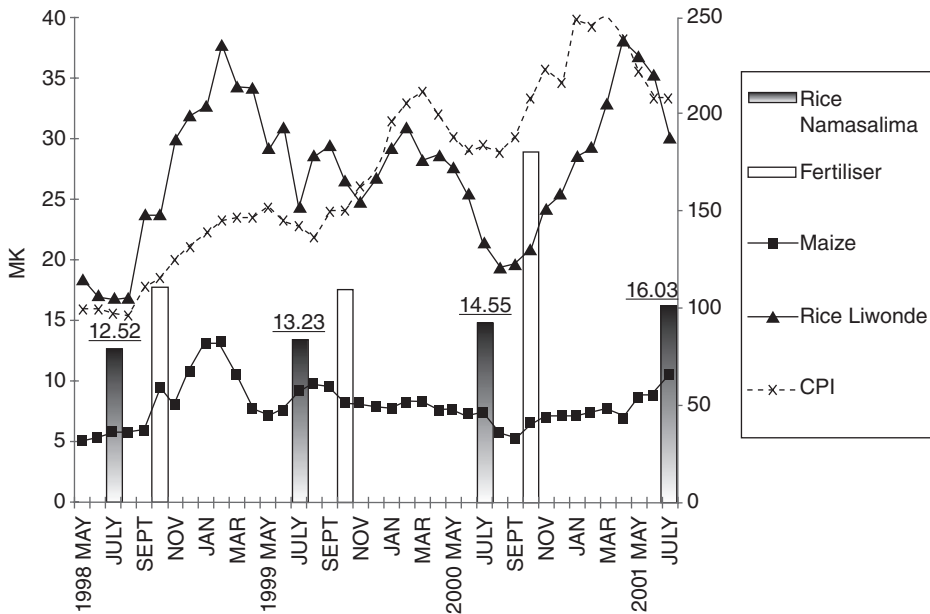
This section has provided evidence of a serious shock to the area in early 2002. Data from various institutions, the village survey and the regional survey of mortality, as well as from qualitative interviews, demonstrate that the crisis was unprecedented in the recent history of the village. Based on this evidence and on several benchmarks, the crisis does qualify as a *famine*. The following section investigates the social and economic conditions in the area before the famine.

PRE-FAMINE CONDITIONS

It has long been debated whether a famine evolves gradually over time, as a 'socio-economic process which causes the accelerated destitution of the most vulnerable, marginal and least-powerful groups in a community' (Walker 1989, 6), or whether a famine is a discrete event, 'a particularly virulent manifestation of [starvation] causing widespread death' (Sen 1981, 40). In order to evaluate the relevance of these propositions, trends in the village before the famine will be examined. The focus is on prices and terms of trade ratios (and production figures) of commodities accounting for a high proportion of household income and expenditure; namely, rice and *ganyu* for income, and maize and fertiliser for expenditure.

The nearest market for which official monthly prices of maize, polished rice and cassava are available is in Liwonde, located about 70 km from Namasalima village (of which about 50 km is on tarmac). No price data covering the period during or before the famine are available from Namasalima itself. Therefore, the Liwonde price data for these commodities are used as proxies for prices in the village. Although there is no evidence on how reliable these proxies are, given the dominant role of ADMARC in maize trade, it is unlikely that substantial differences would exist in the consumer price of maize; the consumer price of rice in the village may, however, vary more significantly from the prices recorded in Liwonde. As displayed earlier, household income in this village is determined by the farm-gate price of rice, which is officially unrecorded. These farm-gate prices are negotiated between traders and producers individually and informally. In the tabulations below, the producer price of rice was computed from the household questionnaire, which collected information from each household about the amount of rice harvested and sold during harvest over the past five years. Sale prices were averaged across the sample and subsequently verified in key informant interviews and in focus group

Figure 4 Trends in prices of dominant commodities before the famine: rice prices for the village are reported in per kilogram units, and take into account milling and post-harvest losses (32 per cent for polished rice) – the ADMARC price of maize was MK5 per kilogram during the period



discussions. Unfortunately, the rice price data from the village refer only to prices at harvest time (July) and do not reflect intra-year rice price fluctuations.¹³

Figure 4 shows the monthly consumer prices of rice and maize at Liwonde market, the producer price of rice in the village during harvest and the price of fertiliser before planting. The prices reported are nominal prices, although the Consumer Price Index (CPI) is also presented in the figure. The main reason for reporting nominal prices instead of prices deflated by the CPI is that the CPI only includes the prices of rice and maize to a limited extent, giving them a total weight of 28.2 per cent in southern rural Malawi. These two commodities are, however, much more important for addressing the concerns of this research, because (1) rice accounts for a high proportion of household income, and (2) the primary concern of this research is to analyse access to maize by the poorest during the famine. In Malawi, studies have estimated that in rural areas more than 80 per cent of the expenditure of the poorest households is on food (82.1, 80.9 and 69.7 per cent by the ultra-poor, the poor and the non-poor households, respectively; see National Statistical Office 1998b, 81, table 36). In addition, the poorest in these areas also receive a higher proportion of their income in kind.

As the majority of households derive a high proportion of their income from selling rice, the price of rice offered to producers at harvest can be used as a good proxy for household wealth. Between 1998 and 2001, the nominal producer price of rice at harvest increased on average by

¹³ As the bulk of the production is sold after harvest and household income is strongly concentrated in rice sales, it is also likely that farmers are able to remember the price at harvest time relatively accurately. Households who did not manage to recall the total quantity of rice produced or the amount of rice sold were excluded from the sample, which then amounted to 130 households; in other words, more than 50 per cent of the village population.

30 per cent, from MK12.5 per kilo in 1998 to MK16 in 2001. While this appears a rather slow but steady increase, it may be concluded that long-term trends in the producer price of rice do *not* predict a gradually developing crisis over the years. The data do, however, suggest a great deal of uncertainty about the functioning of the rice market over the period. While the consumer price tends to follow a certain pattern – increasing after harvest, and peaking during the ‘hungry months’ between December and March before falling at harvest – it is never obvious at what level (and in which month) the price is going to peak: in 1998 and 1999, the rice price peaked in January (although at different levels), while the following year, although prices in January were the same as the year before, they then increased further over the following three months until April. The rice market is therefore rather unpredictable. The unpredictability of *future* prices makes it difficult to schedule rice sales both between harvests and over the long run, which is a severe source of vulnerability for rice producers in the village.

It is possible to investigate whether the increase in the nominal prices of rice has resulted in increasing wealth for rice producers by comparing the producer prices of rice (income) with the prices of other commodities, which account for a high proportion of household expenditure for the majority of households in the village. Table 2 shows ratios between the producer prices of rice, as the major source of income, and the most important sources of household expenditure, such as: (1) the prices of maize; (2) the prices of fertiliser; (3) the consumer prices of rice (at Liwonde, which is used as a proxy for prices of rice in the village if households try to purchase rice during the hungry period); and (4) the prices of cassava. The table shows that the ratio between the consumer prices of rice at Liwonde market and the producer prices of rice in Namasalima village during harvest fluctuated quite intensively over the period of four years. The consumer price was 35, 85, 48 and 88 per cent higher than the producer price during harvest time between 1998 and 2001, respectively. This lack of consistency shows that there is no clear indication on what determines the average producer price of rice in the village during harvest time (or the other way round – the consumer price of rice at the market). The following sentence was echoed in one focus group discussion and confirmed in several other interviews: ‘We are in the hands of traders! They pay for our rice very little money and then they do their own business when they sell it somewhere else. They can make huge profits and we know that they exploit us. But what can we do?’

The free market prices of maize during harvest time show relatively little fluctuation over the period. The average terms of trade between the producer prices of rice and the prices of maize during harvest was 58 per cent over the period, fluctuating between 47 and 69 per cent.

Table 2. The ratio between the producer price of rice in Namasalima (income) and various sources of expenditure, 1998–2001

	1998	1999	2000	2001
Rice at Liwonde	135	185	148	188
Maize at Liwonde	47	69	51	65
Fertiliser	119	125	176	161
Cassava at Liwonde	25	46	30	43

Units in percentages; prices in July. The numbers show the ratio between four major sources of expenditure and the producer price of (polished) rice in Namasalima, as the most important source of income for the majority of households in the village. For example, the table shows that the price of rice at Liwonde market in July 1998 was 35 per cent higher than the producer price of rice in Namasalima during the same month.

As this relatively stable ratio in 2001 (65 per cent) falls within the 'normal and expected range' for rice producers, the prices did not, at this point, predict an approaching famine or raise concern among villagers about the future prices of maize. There was no incentive to keep more rice for future consumption. A similar trend can be observed in the prices of cassava as well, which shows similar (moderate) fluctuation over the years, but nothing exceptional shortly before the crisis.

The price ratio with fertiliser shows a slightly different picture. The prices of fertiliser in 1998 and 1999 were identical, and only 19 and 25 per cent higher than the producer price of rice in the village. This relatively favourable price of fertiliser (compared to the producer price of rice) and the free fertiliser distributed through the Targeted Inputs Programme (TIP) contributed to two bumper harvests in the area.¹⁴ In the year before the famine, however, the price of fertiliser suddenly increased by 65 per cent. This was a much higher increase than the increase in the producer prices of rice (income) over the same period; therefore, the terms of trade between the prices of rice and the prices of fertiliser deteriorated. At the same time, the TIP was scaled down, so access to fertiliser before the famine further declined for the poorest. Whether this could be considered a major causal factor of the famine will be revisited in the section discussing the extent of food availability decline during the famine.

So far, the focus has been on expenditure (maize and fertiliser), but it is important to investigate trends in household income. Given the previously discussed methodological problems, the questionnaire could not capture long-term payments for *ganyu* directly. Sixty-nine per cent of households in the village, however, reported that more work had to be done for the same amount of payment over the four years preceding the famine and only 4 per cent said less; for the remaining households, it was considered the same. This is an indication that, for the majority of households in the village, *ganyu* payments did not keep pace with increases in the price of maize and in the CPI, although no precise estimates could be made.

The data presented above can be interpreted in different ways, and it is difficult to analyse the changes in the standard of living over the period. Certain trends in the patterns of expenditure and in the sources of income might suggest a declining trend in the standard of living of rice producers. While the producer price of rice, the major source of income, increased slightly in nominal terms over the period, payments for casual labour, another important source of income for the poorest, appear to have declined over the years, although the extent of this decline is difficult to quantify. On the expenditure side, the prices of maize and cassava, relative to the farm-gate prices of rice, do not indicate a gradually declining trend, but the terms of trade with the prices of fertiliser deteriorated quite substantially before the famine, which resulted in restricted access to fertiliser. One conclusion that can be derived from the above analysis is that none of the price ratios suggest that the crisis could have been anticipated or that it developed gradually over time. During the harvest in May 2001, before the famine, none of the prices indicated an approaching famine or provided a 'warning' that households should retain more rice or maize for future consumption in the following months. It is therefore

¹⁴ Since 1998, as part of the national food security policy, the government has implemented a free inputs programme through which improved maize seed, legumes and fertiliser have been distributed to smallholder households in Malawi. The Starter Pack initiative (launched in 1998 and then renamed as the Targeted Inputs Programme [TIP] in 2000) had the aims of boosting production and improving food security. During the first two years of the Starter Pack, the programme aimed for universal coverage and targeted 2.8 million smallholders in the country. In the TIP form, it was subsequently scaled down: in 2000/01 only 1.5 million packs were distributed and in 2001/02, during the famine, only one million. The extent to which the scaling down of the Starter Pack contributed to the famine remains an open question; however, 'undoubtedly, the move to TIP played a part' (Harrigan 2008, 246).

necessary to investigate the causes of the crisis from the events that immediately preceded the famine in the village.

CAUSES OF FAMINE

The previous section demonstrated the difficulties with explaining the famine in terms of changes in prices over several years. Even during the harvest in 2001, just a few months before the famine happened, there was no indication from the price data and from the terms of trade between the most important commodities that a famine could be expected to happen. This corresponds with focus group and key informant interviews, where it was frequently mentioned that 'we could not predict what was going to happen'. The crisis, therefore, must have happened as a *shock*, either in production or in access to food. In this section, accordingly, we use the Food Availability Decline and the Food Entitlement Decline theories to investigate the causes of the famine in the village.

Famine as Decline in Food Availability

Food availability can be interpreted in several ways. Before analysing the extent of food availability decline during the famine, two questions need to be clarified. First, which crops should be included in the concept of 'food' (a methodological problem); and second, how reliable are the various production figures (a technical problem)?

The simplest method is to analyse trends in the production of maize, the staple food crop. It is, however, a valid argument that edible 'food' is more than just maize; therefore, the production (and availability) of *other* food crops should also be taken into consideration. As we begin to expand the model, several questions arise: How many (and which) crops should be taken into account? What is the benchmark with which we should compare aggregate food production; in other words, how do we decide if production is sufficient or not? A comprehensive view can be gained by calculating a food balance sheet, which compares food availability with food requirements and identifies the regional 'food gap'. The advantage of food balance sheets is that they make it possible to define food availability in various ways by selecting the mix of crops that are included in the model. They also take into account population growth, thereby providing a more accurate picture of changes in per capita food availability when comparing production over several years.

There is no general rule in the number and type of crops to be included in the food balance sheet. The SADC technical handbook acknowledges that 'in an ideal situation it is desirable to include *all* food commodities in the food balances sheet, however, in practice this is near impossible due in part to the diversity of diets and to difficulties involved in collecting data, especially for relatively minor commodities' (Rook 1994, 7; emphasis added). My purpose here is not to evaluate which methodology is more appropriate but, rather, to demonstrate, for the purpose of the subsequent analysis, the different options to define and to measure food availability. Ideally, a historical food balance sheet at the national level would also take into account the imports, exports and stocks of food commodities, but these cannot be measured accurately at the sub-national level. Therefore, in the following calculations that refer to the 'micro-/regional level', the movement of food commodities will be excluded. Instead, the purpose of the micro-level food balance sheet will be to reveal the *relative* importance of food availability decline during the famine, by comparing levels of production during previous years.

These methodological problems are not unknown either in the Malawian context or in the analyses of other famines. In 2002, for example, the FAO/WFP Crop and Food Supply

Assessment Mission (FAO and WFP 2002) calculated a *cereal* balance sheet (including only maize, rice, sorghum, millet and wheat), while the Famine Early Warning System (FEWS), in addition to these cereals, included cassava, sweet and Irish potato in their calculations (which resulted in a *food* balance sheet) (FEWS 2002). The two models were used simultaneously, although they came up with different estimates of the national food gap. Decline in food availability has been measured in different ways in the analysis of other famines as well. Sen, for example, uses different concepts and measurements of aggregate food availability in his analysis of four famines (Sen 1981). For the analysis of the Bengal famine, an index of 'per capita foodgrains availability' is used, which includes net imports of rice and wheat (*ibid.*, 61). For the Ethiopian famine, food availability refers to 'aggregate production' by districts (*ibid.*, 90) including the production of *all* food crops, but not the import of those crops. For the analysis of the Sahel famine, an index of 'net food output per head' is used, which also includes *all* food crops; and for the analysis of the Bangladesh famine, Sen takes the production of rice, and the availability of 'foodgrains for consumption', including production and net import of wheat as well (*ibid.*, 138).¹⁵ In Sen's analysis of these historical famines, he 'had used all the available production statistics, without exception, including the "unadjusted" and "adjusted" figures of output (and current supply)' (Sen 1987, 10).

In addition to such methodological uncertainties, the reliability of production figures is another important question. The argument that justified government and donor inaction during the famine was that the estimated production of root and tuber crops was so high that there was no 'food crisis, just a maize crisis' in the country. This led to the widespread argument that Malawians have 'inflexible food preferences': 'there is [. . .] usually an outcry when maize is in short supply despite the availability of these other crops, as is the case this year' (FEWS, October 2001). After the famine, it was acknowledged by all concerned that the cassava and sweet potato production figures were 'gross overestimates' (FAO and WFP 2002, 7) and there were several readjustments of these data.¹⁶ Nevertheless, these production figures distorted the analyses of food balance sheets and provided an inaccurate picture of food availability in the country during the famine.

Several people had pointed out well before the famine that root and tuber figures were very unreliable. Carr, for example, noted that, according to the official figures provided by the Ministry of Agriculture, fresh root crop production in the 1997/98 season was as high as 376 kg per person per year, meaning that 'every single family in Malawi would have to eat 5 kg of sweet potatoes per day' (Carr 1998, 3). This appears rather unrealistic, either when contrasted with the very high levels of chronic and acute malnutrition in Malawi, and also when compared

¹⁵ In addition to production and import figures, a frequently used indicator of national food availability is per capita calorie consumption, calculated by the FAO, and Sen also refers to these figures in his analysis of the Ethiopian and Sahel famines.

¹⁶ The FAO/WFP Crop and Food Supply Assessment Mission (FAO and WFP 2002) assumed the 'possible over-estimation of the area and the probable over-estimation of yield' and recommended that previously published figures for cassava be divided by 3.5, and those for sweet potato by 1.5. The Ministry of Agriculture (MOA), on the other hand, suggested that cassava production figures should be divided by (a surprisingly exact) 56.3 per cent and sweet potato figures by 58.2 per cent (Government of Malawi 2002a). Another study based on 'probable hybrid variety shares', divided cassava figures by 2.49, by adjusting the average yield from the official 15 to 6 MT ha⁻¹ (Haggblade and Zulu 2003, 31, table 6). Russell Freeman, IMF staff member posted to the National Statistical Office to assess the real GDP and to evaluate the increase in root crop production, put cassava and sweet potato production each at 9 per cent of the official figures (Freeman 1999); that is, he estimated that the official figures should be divided by roughly 10 (Stephen Carr, pers. comm.). All these different estimates indicate a high margin of error, which was further confirmed by the FAO/WFP Crop and Food Supply Assessment Mission one year later, still emphasising that 'yield and production figures should be treated with caution when considering cassava as part of the overall food availability' (FAO and WFP 2003, 8).

with consumption surveys, which indicate very low levels of food consumption (*ibid.*). In my case study, cassava production figures fall very close to the figures quoted by Carr. According to the official figures during the famine, 382 kg of cassava per person was produced in the region (which translates to 1,443 kcal per day, i.e. 66 per cent of the total per capita calorie requirement of 2,200 kcal).¹⁷ In what follows, several different food balance sheets are calculated for the region in which the decline in food availability is interpreted in three ways: (1) the maize gap; (2) the cereal gap; and (3) the food gap including not only cereals, but also (a) cassava and sweet potato, (b) other (non-horticultural crops) and, finally, (c) all crops including fruits and vegetables.

At first, the decline in food availability appears to be a significant cause of the famine. Table 3 shows that *maize production* in 2001 declined by 45 per cent, which is even more significant than the 32 per cent fall at the national level (Devereux 2002). Much of this decline in production was due to excessive and erratic rainfall in the region, as well as localised flooding. An FAO Assessment Report, dated 21 March 2001, concluded that Zomba District 'has been seriously affected by water logging conditions in areas around lake Chirwa and other low lying areas' and the Nsondole Environmental Planning Area (EPA) was singled out as one of the four worst-affected areas in the region (Nastasi and Sande 2001, 8). In addition, the EPA's agricultural officer reported on 6 February 2001 that 'heavy rainfall has caused severe downfall of houses in Namasalima Section' (Government of Malawi 2001, 1). As a consequence of the decline in maize production, the maize gap in the area increased more than five times, from 915 MT to nearly 5,000 MT. Extending the analysis to the *production of all cereals* (maize, rice and sorghum), the decline was 38 per cent, a smaller fall than in the production of maize – but still quite substantial. Between 1998 and 2001, rice production increased more than 2.3 times: while in 1998 only 14 per cent of total cereal production was rice, in 2001 the proportion rose to 34 per cent. As a consequence, a cereal *surplus* of 111 MT in 2000 turned into a massive deficit of 4,182 MT in 2001. Both the production and availability of maize and cereals imply that the decline in food availability was a significant causal factor of the crisis.

Before concluding that the decline in food availability caused the famine, some counter-arguments should be considered. First, 1998 was very similar to 2001 in terms of production of maize and cereals: maize production in 2001 was only 4 per cent lower than in 1998 and, taking into account population growth, the maize gap was 12 per cent higher. The production of *all cereals* in the region, however, was 18 per cent *higher* in 2001 than in 1998, and even if we factor in population growth, the cereal gap was nearly 15 per cent *lower* during the famine than in 1998. The production of rice during the famine, although decreased by 11 per cent compared to the previous year, was 30 per cent higher than in 1999 and 232 per cent higher than in 1998. More interestingly, if we include cassava and sweet potatoes (either by using the original – overestimated – or the adjusted figures), together with other crops and fruits and vegetables, *all* the food balances are *positive*, indicating a food *surplus* for every year over the period, including the famine year.¹⁸

¹⁷ Given the magnitude of overestimation and the very high margins of error of different readjustments of the production figures, it is likely that these arguments were used to justify inaction, and were driven by the lack of political will to intervene during the famine rather than by technical uncertainties. This question, however, goes beyond the scope of this paper.

¹⁸ If we assume that root crop production figures were overestimated by a factor of 10, the highest adjustment factor recommended in Malawi, the calculations indicate a food *deficit* of 4,447 MT for the famine year. This is only 11 per cent lower than the maize deficit of 4,997 MT, indicating that the contribution of cassava and sweet potatoes to food availability becomes negligible. Even in this case, however, the final food balance, including *all* food crops produced in the region, still indicates a food *surplus* of 3,135 MT.

Table 3. (Continued)

Requirement		Crop	1997/98	1998/99	1999/00	2000/01
Deficit/surplus	Population ^c					
	Maize		61,076	61,931	62,798	63,677
	Cereal		9,528	9,661	9,797	9,934
	Food with and without HC		10,658	10,807	10,958	11,112
	Maize		14,212	14,411	14,613	14,818
Food: cereals and . . .	Cereal		-4,379	-440	-915	-4,997
			-4,791	-67	111	-4,182
		Cassava and sweet potato	Not available	8,434	22,633	19,758
				1,519	7,733	4,008
		Excluding horticulture		9,086	23,255	20,433
				2,170	8,354	4,684
		Including horticulture		15,624	30,733	28,016
				8,709	15,833	12,267

Source: Zomba RDP production figures. Data for 1997/98 collected from Nsondole EPA.

^a All production figures are metric tons converted to maize equivalent, including post-harvest losses.

^b Cereals include sorghum, except for the year 1997/98.

^c Food requirement figures are calculated from the FEWS food balance sheet (June 2001). The population figure for 1998 is adopted from the National Statistical Office; the intercensal growth rate of 1.4 per cent (Southern Region) has been used to calculate the annual increase (National Statistical Office 1998a, 9, table 1.2).

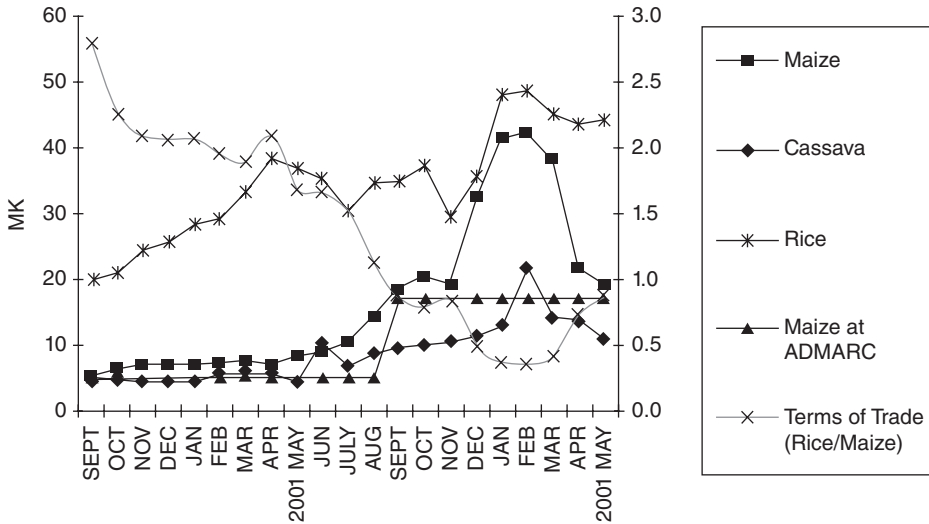
The implications of the above calculations are important, since the conclusions that can be drawn from the different interpretations of the food balance sheet are contradictory. A very high maize (and a high cereal) *gap* turns into a food *surplus* when we include other food crops (and adjust the requirement figures accordingly). However, even if we use the official 'adjusted' figures for cassava and sweet potatoes, and exclude horticultural crops, the *food surplus* remains as high (over +4,000 MT by taking the adjusted cassava and sweet potato figures) as the *cereal gap* (-4,182 MT) during the famine. Due to the very high margin of error of the production figures, as well as the flexibility of the methodology in choosing the mix of crops that could be included in the calculations, it remains challenging to draw conclusions about the extent of the decline in food availability during the famine. However, in spite of these data and methodological uncertainties, one firm conclusion can be drawn from the food balance sheet analysis. In 1998, the production of maize and cereals, as well as the size of the food (maize and cereal) gap, was very similar to the level of production during the 2001/02 famine. Assuming that the margin of error in estimating the production figures was the same over the period, the figures for these two years are comparable. There are no reasons to believe that the area suffered from a food crisis in 1998. Apart from a mild cholera outbreak in 1999, school enrolment and child malnutrition figures remained 'normal'. In none of the qualitative interviews was any sort of crisis mentioned; the fortnightly reports do not indicate a crisis; and no aid programme was implemented in the village. Thus, further analysis is required to understand why a very similar situation in 2001 caused a famine in the region, while there was no crisis in 1998.

Famine as Decline in Food Entitlement

In this section, 'exchange entitlement' is analysed by examining changes in prices, in terms of trade ratios, in incomes and in food markets. Figure 5 shows the rapidly increasing prices of all food crops after the harvest in June 2001. ADMARC raised the price of maize in September from MK5 to MK17 per kilogram, but maize prices continued to increase until February, peaking at MK42, a level more than six times higher than the year before.¹⁹ Prices of rice and cassava also increased during the same period, peaking in February 2002. Cassava cost twice as much in September and 394 per cent more in February than it had cost in the same months the year before, which indicates excess demand for cassava on the market and confirms the fact that the official cassava production figures were overestimates. In addition, the prices of horticultural crops also increased significantly. While, during 'normal times', fruit and vegetables are considered inferior crops – a type of snack eaten to supplement, but not to substitute for, the main meal (*nsima*) – during the famine, these crops changed from their secondary and inferior nature, to be seen as commercial food crops with an increased market value. The prices for fruit were between three to five times their normal level, which is less than the increase in the price of maize (six times), but which does correspond with the increase in the price of cassava (four times). Most of these horticultural crops (especially bananas and mangoes) were often stolen during the famine, a phenomenon that had rarely happened before. This confirms that (1) there was a general lack of *food*, and not only *maize*, in the community, and the crisis

¹⁹ Songani market is directly linked to Liwonde market, which is approximately 60 km to the north. From Liwonde, there is a direct train to Mozambique, where maize was available throughout the crisis, remaining the major source for private traders to import maize and sell at Songani. Information about the prices of maize in Mozambique, as well as profit margins in the maize trade during the famine, could not be obtained from the interviews due to sensitivity of information, and it was beyond the scope of the fieldwork to visit the relevant areas.

Figure 5 The prices of the main food commodities during the famine



was not simply a *maize* crisis but a *food* crisis, a *famine*; and (2) food preferences are indeed ‘flexible’ and in times of need people are willing to purchase, and consume, *any* kind of food crop, including ‘inferior’ crops.

The increasing maize prices had an impact on the terms of trade between the producer prices of rice (income) and the consumer prices of maize (expenditure). Between 1998 and 2000, the terms of trade at harvest in July fluctuated between 215, 146 and 196 per cent, respectively, and in July 2001 the ratio was still within the ‘normal’ range at 153 per cent – that is, it was possible to exchange 1.53 bags of maize for one bag of rice.²⁰ The average terms of trade in 1998/99 and 1999/2000 were 162 per cent and 161 per cent, respectively, while during the year before the famine the ratio was 210 per cent, which was exceptionally *favourable* for rice producers. This, again, is an indication that there was no sign of an approaching crisis by the time rice was being harvested and sold before the famine in July 2001. Maize prices, however, started to increase after the harvest in July 2001. By then, cash-constrained producers had already sold their rice harvest and the terms of trade collapsed to 38 per cent in February 2002.

As noted earlier, an important source of income for the poorest people is *ganyu*. Even in ‘normal years’, the poorest can only survive on the basis of the income they derive from casual labour during the hungry months. The relatively good harvest of rice in the area and at the Domasi Scheme in 2002 implies that the normal levels of labour inputs were used and that aggregate demand for agricultural labour did not decline during the famine. According to the questionnaire, 83 per cent of those who did casual labour during the crisis worked in rice fields, most of them in January (26 per cent) and in February (62 per cent). The local labour market,

²⁰ If we take the price of maize at ADMARC for the calculations, we get an even more favourable terms of trade ratio for rice producers: before the famine it would have stayed at 291 per cent (constant – as ADMARC prices are fixed), rising to 321 per cent in July, only few weeks before the famine, but suddenly dropping to 94 per cent as ADMARC raised the price of maize to MK17. The ratio would have stayed at this level had ADMARC had maize during the famine – which would still be below average, but perhaps not so devastating as 38 per cent.

however, was obviously affected by the declining terms of trade, and came under pressure from both the demand and the supply side. The more cash-constrained and less wealthy 'average' rice producers, who would in 'normal times' employ some *ganyu* to substitute for their own labour, were negatively affected by the collapsing terms of trade and had no further resources to employ wage labourers. The magnitude of this decline in the effective demand for labour was so great that the one large estate in the area was unable to absorb the underutilised labour. The owner of the estate noted, 'it was impossible for me to provide *ganyu* to everyone who came to me during the hungry months. I can always employ people, but in January and February I had to send people away, to whom I would normally give work.' On the other hand, supply of labour on offer from the poorest households, who were desperate for work, was increasing. In the questionnaire, 52 per cent of households reported that at least one of their members had searched for, but could not find, employment during the crisis. The qualitative impact of the changes in the local labour market was that the stronger and more productive workers found employment, while the weak and less productive (mainly the elderly) members of the labour force had less chance of getting work: in 73 per cent of the cases, it was the head of the household who did *ganyu*.

The real value of payments for *ganyu*, which can be made in kind or in cash (Englund 1999; Bryceson 2006), declined at this critical juncture. During the famine, according to the household questionnaire, in 90 per cent of the households where at least one household member did *ganyu*, these workers were paid in *cash*. This finding contradicts the observation that 'in the 2001–02 main agricultural season [. . .] *ganyu food* payments were the main means of obtaining food' (Bryceson 2006, 186; emphasis added). The average nominal wage rate during the famine remained the same as in 'normal times'. During early 2001, before the crisis, an NGO paid a daily wage of MK20 on a road construction project, for six hours of work. This is the same as the rate paid by the Malawi Social Action Fund (MASAF) in July 2002, after the crisis, on a dyke construction project in the nearby villages. During the famine, the real value of this wage expressed in terms of maize collapsed. With the prevailing maize prices in January and February (MK42.3), the average daily payment was sufficient to purchase 0.47 kg of maize, which, for an average household containing 4.1 people in the village, is equivalent to 0.115 kg (396.7 kcal) of maize per day per person, or 18 per cent of the average daily per capita energy requirement.²¹ The same amount of payment during the same month in the previous year would have been sufficient to buy 4 kg of maize (MK5 per kilogram at ADMARC), 8.5 times as much as during the famine.

The crucial factor that triggered the collapse in the terms of trade and indirectly adverse labour market consequences was the dramatic increase in the price of maize, which could have been prevented by a successful intervention in the market by ADMARC. It is well recognised in Malawi that the operations of the government marketing board support the poor both directly, by providing access to maize at affordable prices and maintaining terms of trade between commodities, and indirectly, by sustaining demand and supply in the local labour market, thereby maintaining the normal functioning of the rural economy: 'When ADMARC markets have maize in stock, they become the most reliable and affordable source of maize for rural households, particularly the *poor* and food insecure, . . . and the *rich*, who buy the maize to pay "ganyu" workers or tenants' (Mvula et al. 2003; emphasis added).

ADMARC had historically played a significant role in regulating maize prices. It was able to intervene successfully in 1998, when the level of maize production and maize and cereal

²¹ The calorie content of maize is calculated by using 3,450 kcal kg⁻¹ and the average per capita calorie requirement is 2200 kcal (FAO, WHO and UNU 1985).

deficits were of a similar order of magnitude, thereby preventing a major crisis. During 1998, maize prices on the free market increased to MK13 and terms of trade with rice fell to 96 per cent in February, a less significant fall than the MK42 price of maize and the 38 per cent terms of trade decline in February 2002. In 1998, however, the local branch of ADMARC sold 245 MT of maize between August and February, equivalent to the maximum storage capacity of the depot (250 MT). This was a massive intervention, certainly when compared to the interventions in 1995/96 and 1996/97, when only 16 and 10 MT of maize were sold by ADMARC between December and February, respectively. Unfortunately, no data for maize sales from the local ADMARC depot are available for the year 2001/02, but it is known that the year before the famine – even during the bumper harvest – ADMARC sold as much as 76 MT of maize, nearly one third of its storage capacity, between November and January. It is thus reasonable to conclude that there *was* a high demand for maize at ADMARC. The household questionnaire indicates that 84 per cent of the households failed to purchase maize at ADMARC during the crisis, which confirms that ‘ADMARC markets were characterized by long queues of starving people who sometimes spent nights at the markets but failed to buy maize in the end . . .’ (Mvula et al. 2003, 33). The importance of price regulation for food security is highlighted by the fact that 87 per cent of the households responded in the questionnaire that if ADMARC had had maize during the famine, there would not have been a famine in the village.²² This study therefore provides evidence of the importance of intervention and of regulation of the price of maize, because the price increase in this ‘strategic’ commodity trickles through the entire economic system and affects all sectors of the local economy, triggering an increase in the prices of other food commodities, leading to a decline in real income of villagers, negatively impacting the local labour market, and disrupting coping and survival strategies.

One of the central arguments for reducing the role of ADMARC in the market is that the private trade system would then emerge and take over its role in rural areas (Mvula 2003). The validity of this argument has been hotly debated on both theoretical and empirical grounds. The findings of this case study suggest that the private market at the local level *did* take over the role of ADMARC, which can be demonstrated by analysing the ‘capacity’ of local private traders (measured by the number of vendors in the area and their ability to operate in the region) and their economic incentives. For example, based on the assumption that ADMARC would have sold 250 MT of maize during the famine – that is, the same amount that they sold during 1997/98 – and that one small-scale trader can transport one 50 kg bag of maize per day by bicycle from the tarmac road, it would require 55 vendors to import the amount of maize that would have been sold by ADMARC during the three worst months of the crisis.²³ The assumption appears reasonable if we accept that the average vendor in Malawi operates within

²² It is interesting to note that no data were available about ADMARC maize sales during the famine. This is strange, considering that some of the figures for earlier years were available, and it could be considered a sign of corruption. There were clear incentives for ADMARC officers and truck drivers to profiteer from the difference between the subsidised ADMARC price of MK17 and the free market price of maize, which reached MK42 between January and March 2002: ‘The short supply of maize led to bribery and nepotism in the selling of the available maize at some ADMARC markets . . . In some places, ADMARC officers were reportedly serving people from the surrounding villages first or vendors . . .’ (Mvula 2003, 33).

²³ To import 250 MT of maize in three months’ time would require 55 small-scale vendors (= 250 MT per 50 kg bag per 90 days). The population of the Namasalima village group (13 villages) in 2003 was 5,892 people, of which 32.3 per cent (1,903 people) would have been between the age of 20 and 45 (National Statistical Office 1998a, 29, table 3.2). If we take the catchment area of the Health Centre (43 villages), it would be sufficient if slightly more than one person per village would engage in trading. Transport between the tarmac and the village by bicycle cost MK130 during the fieldwork. The price of (a 50 kg bag of) maize was MK250 before the crisis (ADMARC price) and MK2,000 at the height of the crisis (prices at Liwonde market).

a radius of 53 km (Mvula 2003, 61); that the village is only 16 km from the tarmac; and that it would take less than 3 per cent of the working-age population of the village group, consisting of about 13 villages strongly reliant on ADMARC for maize purchases, to engage in trading activities for a period of three months. Furthermore, the major disincentive faced by private traders operating in remote villages has become far less significant as maize prices have increased: transport costs as a proportion of the total price of maize declined from 34 per cent to 6 per cent during the famine. This argument was confirmed in the key informant interviews, which revealed that the price of maize in the village market was *not* substantially higher than at the tarmac, and that there *were* traders transporting maize to the village during the crisis at prices similar to those at the main market. While in 1998 'private traders were hampered by credit constraints and did not move into the remote markets abandoned by ADMARC' (Harrigan 2003, 850), in 2001, in this particular area, private traders did transport maize from the tarmac to the village. Therefore, the ultimate reason why the majority of the village population was unable to access maize was neither the unavailability of maize in the area nor the inability of the local small-scale private trade system to transport maize to the village. Most households in the village did not have the purchasing power to afford to buy maize at the local market due to the very high prices. This increase in the price originated from the national level.

The above discussion raises interesting questions about the definition of 'private trade' and its potential to satisfy the pattern of demand for maize in this area. Under the liberalization agenda of the 1980s and 1990s, much attention has been devoted to the relative inefficiency of public trade compared with private trade *in general*, but less attention has been focused on a *disaggregated* analysis of private trade according to size of the agents involved. The fact that the private trade system was functioning during the famine in this area implies that the sub-national level is insufficient to analyse the *causes* of the market failure; only its *consequences* can be analysed at this level. Market prices across the major markets in the country were moving together during the famine (FEWS June–July Report 2002, 4), as is confirmed by the stable coefficient of variation of maize prices over the period (Tiba 2005, 92). What matters, therefore, is less the functioning of the private trade system at the local or sub-national level but, rather, the *large institutional orders* of maize, which did not arrive during the crisis. The interlinking local markets are determined by the functioning of the national market. Just as the village market depends on ADMARC, and subsequently on the nearest of the major local markets in the country, these markets also depend, in turn, on institutional orders placed by the government, by *large* private trade companies and/or by aid agencies. It may, therefore, be insufficient to talk about 'private grain traders' *in general*, who 'hoarded supplies and made money out of food shortages' (Owusu and Ng'ambi 2002, 11). At the micro-level these agents indeed, rationally, 'followed the market signals all too well' (*ibid.*) and may have profiteered from the crisis through the increased profit margins; however, such agents are unlikely to have the capacity to hoard large quantities over time. This, if true, is more likely to have happened at the national level, among the larger institutions involved in maize trade.

Finally, the evolution of the famine and the failure of ADMARC to intervene are well documented in the fortnightly reports of the Nsondole EPA, which provide evidence that crucial information was 'concealed, ignored, or mismanaged' (Devereux 2002, 5) during the famine. While the report of 10 April 2001 still concluded that the food situation was 'adequate', with 'no coping mechanisms', the report of 3 September warned that 'the area is experiencing shortage of food', which, from this point in time, was confirmed in *all* subsequent fortnightly reports until the harvest of May 2002. On 13 November 2001, the agricultural officer reported that 'maize is being bought far from the EPA, about 15–20 km away' and said that Namasalima group village was among the most food-insecure areas. On 11 December, the report repeatedly

noted that ‘shortage of food in the area continues, maize is bought *very* far from the EPA’ (emphasis added), while on 9 January 2002 it concluded that ‘there are no major markets in the area be it ADMARC or any major trader’ and only Zomba and Liwonde were considered as ‘functioning markets’, these being 35 km and 65 km away from Namasalima village, respectively. The report of 18 February simply stated: ‘maize is *not* available in the near markets’ (Government of Malawi 2001–2002). This sequence of reports confirms that the early warning system in the rural areas was functioning appropriately, and there was sufficient and reliable information available at the local level. On the other hand, this contradicts the findings of various national-level analyses. For example, a Joint Multi-Donor Survey conducted in November 2001 concluded that ‘local and ADMARC markets are well supplied with sufficient quantities of maize to meet localised demand’ (EU, FEWS and WFP 2001) and the Famine Early Warning System two months later also reported that ‘most local markets and some ADMARC markets have sufficient maize from various sources’ (FEWS, January 2002 report, 4). Devereux regards these reports as ‘at best misleading, at worst factually incorrect’ (Devereux 2002, 16), and this case study further confirms that the information recorded at the local level was not transmitted to the national level. Warnings of an accurately documented and rapidly evolving famine were ignored, which raises the hypothesis that information from the field was politicised at the national level.

CONCLUSION

Several questions about Malawi’s 2001/02 famine still remain unanswered. This paper has attempted to contribute to the debate with a case study from a village. The research questions focused on the severity, impact and causes of the crisis, each of which has been subject to debate in macro-level analyses. Extrapolating from a micro-case study to the national level always poses conceptual and methodological problems, but grassroots research can provide a more authentic view of how people see and experience crises.

Several indicators show the severity of the crisis at the village level. Malnutrition, the frequency of hunger- and non-hunger-related diseases, school enrolment and mortality trends, as well as qualitative interviews, all indicate a severe shock to the community. Compared with conventional benchmarks, the crisis does qualify as a famine. As for the causes of the crisis, not enough evidence has been found to support the view that long-term trends *caused* the shock. Price ratios of important crops that account for a high proportion of household income and expenditure fluctuated over the period, but there was no clear trend suggesting that the community was ‘sliding’ into the crisis. The production figures give an ambiguous picture: a severe decline in maize production and an increase in the maize gap was partly offset by cereal production, and completely compensated for by root and tuber production, as well as by the production of other food crops. While the reliability of these figures is open to question, a comparison with 1998 clearly shows that the major cause of the crisis could not have been production decline alone.

It is much easier to explain the onset of the famine by examining maize prices, which increased exponentially and rapidly during the period. The maize prices altered the terms of trade ratios (since the rice producers had already sold their harvest, as they were unable to foresee the approaching increase in maize prices), impacted the local *ganyu* market, which was affected by the increased supply of, and declining demand for, labour, and indirectly triggered further increases in the prices of other food crops, including rice, cassava and fruits. According to most of the villagers, they would have been able to cope with the crisis had ADMARC offered to sell maize at affordable prices during the famine.

While recognising factors including the multiple and overlapping causes of the crisis, the methodological problems involved in measuring food security at the household and individual level, and the unreliability of the data, this paper has demonstrated that the principal causal factor precipitating the crisis was the unexpected price increase of maize during the year 2001/02. Although there may have been a decline in local food production, if prices had not surged so rapidly and unexpectedly, there would not have been a crisis. The impending crisis was well and accurately documented at the local level, but this information was not transmitted to the national level. An explanation for this failure demands a national-level enquiry.

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